Format WPR

AMITY SCHOOL OF ENGINEERING & TECHNOLOGY
B.Tech (CSE - AIML) VII Semester

Project Area -- Generative AI & AI Safety (Trustworthy AI / MLOps)

Students Weekly Progress Report (WPR — Week 12) For Odd Semester of session 2026-2027

To be filled by Students	Group No. 50
Students Name	1. Shourya Solanki 2. Rachit Ryan Chug 3. Dhruv Raj Singh
Roll no.	1. 01 2. 02 3. 03
Enrollment no.	1. A2305223569 2. A2305223166 3. A2305223191
Project Title finalized, if Yes, give name, if NO, give reason	Yes. Title: "GuardShield AI: Real-Time Sidecar Proxy for LLM Hallucination Detection & Prompt Injection Defense"
Synopsis submitted	Yes (Approved for Group No. 50)
Literature review	Synthesized contemporary AI safety and prompt injection guard literature (September–October 2026 preprints): Liu et al. on benchmark contamination (arXiv:2610.03448), Zhang et al. on paired benign-twin controls (arXiv:2609.26176), HiveTraceGuard-Pro on recall/FPR trade-offs (arXiv:2609.01046), and PIDS-Bench. Formulated the Paired Joint Accuracy (PJA) evaluation metric and executed an exhaustive paired audit across 17 open-weight and commercial prompt-injection detectors and safety SLMs, demonstrating format-triggered over-defense and tokenizer fragmentation bias.
Technical & Economical Feasibility	Feasible. Dual-tier evaluation framework executed across local NVIDIA RTX 3050 GPU (4GB VRAM) for low-latency inference (<30ms Stage 1, <31ms Stage 2) and cloud-accelerated Kaggle/Colab environments (NVIDIA T4, L4, A100 GPUs) for multi-model parallel evaluation (17 models, 12,000+ evaluated pairs). Open-source software stack with zero licensing expenditure (₹0).
Bill of Material	₹0.00 (Fully open-source software stack, public academic benchmark datasets, Hugging Face Hub open-weight models, and local developer environment)
Project Progress Schedule (PERT Chart)	• W1-W5: Problem formulation, dataset curation, DeBERTa fine-tuning & evaluation (Completed) • W6-W8: Streaming entropy engine, NLI fact verifier, and multi-model benchmarking (Completed) • W9-W10: Stage 2 authentic benchmark integration (FEVER + HaluEval), NLI declarative statements, 95% Wilson CIs, and E2E latency measurement (Completed)

	• W11: Interactive unified pipeline with multi-model backbone switching, negative refutation hallucination resolution, 15k massive multi-domain dataset curation, and high-VRAM DeBERTa-v3-large trainer (Completed) • W12: Comprehensive 17-guard paired perturbation audit (Study A: 12 families, 500 pairs), native Hindi/Hinglish benchmark curation (Study B: 400 seeds), pre-registered research spec (SPEC.md), novelty gap analysis against 2026 preprints, and Paper 1 manuscript preparation for IEEE/ACM submission (Completed / Week 12)
Design of critical components	1. Paired-Perturbation Audit Harness & PJA Metric Engine (guard_audit/perturbations.py & paper1/SPEC.md) 2. Native Multilingual & Hinglish Diagnostic Benchmark Suite (paper1/annotation/seeds.csv & validate_seeds.py) 3. Multi-Guard Evaluation Engine for 17 SOTA Defenses (paper1/runs/obf_main/ across Llama-Guard-3, Prompt-Guard-2, ProtectAI, Qwen3Guard, Sentinel-v2, WildGuard) 4. Cryptographic HMAC-SHA256 Signed Assistant History Proxy (src/proxy_server.py) 5. Automated LaTeX Table & Manuscript Compilation Pipeline (scripts/generate_readme_tables.py & paper1/runs/obf_main/)
Fabrication work (give %)	0% (Software Project)
Experimental work (give %)	100% (Stage 1, Stage 2, and 17-Guard comparative audit completed; 12 paired perturbation families evaluated; 500 pairs scored per guard; multi-turn signed history verified with 0% false block rate)
Result and Analysis	Demonstrated that attack-only evaluations are fundamentally misleading: under character obfuscation (zero-width, soft-hyphen, leetspeak, homoglyphs), small classifiers (Deepset, FMOps, Mezzo) suffer blanket fail-closed collapse (FPR rises to 79.4%–100.0% on benign twins), masquerading as high attack recall. Horizon Guard Base and Qwen3Guard-8B maintained superior Paired Joint Accuracy (PJA 93.8%–98.8%). GuardShield Stage 1 with HMAC assistant signing achieved 0.00% conversation block rate on multi-turn dialogue (vs 20.00% for ProtectAI) and 3.08% FPR on hard benigns with 60.52 ms median E2E latency.
Report writing	WPR Week 12 drafted, verified with authoritative empirical audit logs across 17 guards, and formatted.
Signature of students	1. Shourya Solanki 2. Rachit Ryan Chug 3. Dhruv Raj Singh

Work done in this week (Week 12):
1. Paper 1 Formal Formulation, Hypotheses Pre-Registration & Landscape Review: Formulated and registered the formal experimental specification (`paper1/SPEC.md`) titled "Guards that reject the unfamiliar: format-triggered over-defense in prompt-injection detectors". Conducted exhaustive novelty verification (`reports/Guardrail paper novelty check.md` and `reports/LLM guardrail research gaps.md`) against late-2026 preprints (Liu arXiv:2610.03448, Zhang arXiv:2609.26176, HiveTraceGuard arXiv:2609.01046, PIDS-Bench). Pre-registered core hypotheses (H1–H5), establishing the Paired Joint Accuracy (PJA) evaluation metric where both an attack and its length-matched benign twin must be classified correctly under identical, label-blind transformations.
2. Large-Scale Paired Obfuscation Audit Across 17 SOTA Prompt Injection Guards (Study A): Engineered and executed a comprehensive multi-model audit harness evaluating 17 guardrails and safety SLMs: GuardShield DeBERTa-v3,

ProtectAI v2, Meta Prompt Guard 1 (classifier + jailbreak), Meta Prompt Guard 2 (22M & 86M), Meta Llama Guard 3 (1B & 8B), Deepset, FMOps, Horizon Guard Base, Mezzo v2 Base, Perplexity GPT-2, Qwen3Guard (0.6B & 8B), Sentinel v2, and WildGuard. Tested 500 attack-benign pairs (1,000 prompts per condition) across 12 perturbation families: `clean`, `zero_width`, `soft_hyphen`, `leetspeak`, `homoglyph`, `base64_wrap`, `base64_scaffold`, `rot13_wrap`, `rot13_scaffold`, `fullwidth`, `diacritics`, and `spaced_letters`.
3. Empirical Discovery of Format Over-Defense, Fail-Closed Bias & Scaffold Controls: Uncovered that traditional attack-only recall benchmarks are severely confounded: under zero-width spaces, soft-hyphens, and homoglyphs, baseline classifiers (Deepset, FMOps, Mezzo) exhibit near 100% attack recall only because their benign twin FPR surges to 79.4%–100.0% (blanket fail-closed rejection). Introduced label-blind scaffold controls (`base64_scaffold`, `rot13_scaffold`) with identical plain-text phrasing, proving that wrapper rejections are driven primarily by prompt template phrasing rather than base64 decoding ability. Tracked token-length truncation profiles across all guards to prevent unfair evaluation.
4. Native Multilingual & Hinglish Diagnostic Benchmark Curation (Study B): Designed and curated a 400-seed diagnostic evaluation benchmark (`paper1/annotation/seeds.csv`) covering English (`en`), Devanagari Hindi (`hi_deva`), Romanized Hindi (`hi_roman`), and Hindi-English code-mixing (`hinglish`) across 5 critical injection categories and benign everyday tasks (30% trigger-word split). Authored strict annotation guidelines (`paper1/annotation/GUIDELINES.md`) mandating native-speaker composition (excluding machine translation artifacts) and developed an automated seed validation tool (`paper1/annotation/validate_seeds.py`) enforcing register consistency and Cohen's κ label checks.
5. Cloud-Scale Kaggle / Colab Automated Benchmarking Architecture: Developed turnkey Kaggle and Google Colab execution environments (`paper1/NOTEBOOK_SETUP.md`) supporting single-pass verdict scoring across NVIDIA T4, L4, A100, and TPU v5e accelerators. Implemented exponential backoff retry mechanisms for Hugging Face Hub API rate limits, dynamic chunk-size adaptations for generative guards (Llama Guard 3, Qwen3Guard, WildGuard), and automated JSON/JSONL score serialization with SHA-256 dataset manifests.
6. Rigorous Audit Remediation, HMAC Assistant Signing & Production Hardening: Resolved all independent technical auditor findings (Re-Audit #6 and #7). Deployed cryptographic HMAC-SHA256 assistant response signing in `src/proxy_server.py`, ensuring authentic multi-turn LLM replies are recognized without scanning while keeping user inputs and forged assistant turns under unconditional neural inspection. Achieved 0.00% conversation false blocks on multi-turn dialogue (vs. 20.00% for ProtectAI) and reduced benign-hard FPR to 3.08%, with clean git commit synchronization and verified packaging under 500 MB.

To be filled by Guide (strike off whichever is not applicable)
Performance of students is satisfactory
Performance of students is unsatisfactory
A warning to be issued to student(s) (Name)
Student was not well (Name)
Date Signature of Guide